CASE STUDY – ASSIGNMENT TOOL 2

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1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behaviour.

Case Study

A warehouse robot has to pick, pack and transport items while avoiding obstacles and completing the work quickly.

MDP Components

State:

The state represents the current condition of the robot, such as its location, item location, destination and nearby obstacles.

Actions:

The robot can move forward, backward, left or right, pick an item, place an item and stop.

Rewards:

Correctly picking an item → +5

Delivering an item → +10

Hitting an obstacle → -10

Wrong pick/place → -5

Taking unnecessary time → -1

Policy:

The policy tells the robot which action is best for a particular state.

Environment:

The warehouse, shelves, items, workers, obstacles and delivery locations form the environment.

Analysis

The reward function should balance speed, accuracy and safety. If the collision penalty is high, the robot learns to move safely. If fast delivery receives a good reward, it learns to complete tasks quickly. Therefore, proper reward design helps the robot learn safe and efficient behaviour.

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2. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation.

Case Study

A streaming platform can use RL to recommend movies according to the user's preferences and viewing history.

State:

The state includes user's previously watched movies, preferred genres, ratings and viewing history.

Actions:

The action is recommending a movie or selecting a particular type of movie for the user.

Rewards:

User watches the recommended movie → +5

User gives a good rating → +10

User skips the movie → -2

User dislikes the movie → -5

Exploration vs Exploitation

Exploration means recommending new or different movies to understand the user's interests.

Exploitation means recommending movies similar to the user's already known preferences.

Analysis

If the system only recommends familiar movies, users may get bored. Too much exploration can also result in irrelevant recommendations. Therefore, the RL system should maintain a balance between exploration and exploitation to improve user engagement.

3. An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm.

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Case Study

RL can be used in agriculture to decide the correct amount of irrigation and fertilizer using soil conditions and weather information.

States:

Soil moisture, temperature, weather forecast, crop condition and growth stage.

Actions:

The system can increase/decrease irrigation, increase/decrease fertilizer or take no action.

Rewards:

Good crop growth → +5

High crop yield → +10

Saving water → +3

Excess water/fertilizer → -5

Poor crop growth → -10

Delayed Reward

The result of irrigation or fertilizer may not be visible immediately. Its effect may be observed only after several days or at harvest time. Therefore, the system has to learn from delayed rewards.

Suitable Algorithm

Q-Learning can be used because the agent can learn which actions provide better long-term rewards through repeated interaction with the environment.

Analysis

The RL system can gradually learn how much water and fertilizer are required under different conditions, helping improve crop yield while reducing unnecessary resource usage.

4. A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modelled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making.

Case Study

RL can control lighting, heating and cooling automatically depending on occupancy and weather conditions.

States:

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Room temperature, number of people, outside weather, time and current energy usage.

Actions:

Turn lights ON/OFF, increase/decrease temperature and turn heating or cooling ON/OFF.

Rewards:

Comfortable room condition → +5

Saving electricity → +5

High unnecessary energy usage → -5

Uncomfortable temperature → -5

Comfort vs Energy Saving

There can be a conflict between user comfort and energy saving. For example, continuously running the AC may provide comfort but consume more electricity.

The reward function should therefore consider both objectives.

Reward Shaping

Reward shaping means giving additional rewards or penalties to guide the agent towards desired behaviour. Proper reward shaping can help the smart home learn to maintain comfort while reducing energy consumption.

4. A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies.

Case Study

A gaming company can use RL to train an agent to play a strategy game against human players.

MDP Components

State:

The current game situation, player position, opponent position, available resources, score and remaining moves.

Actions:

Move, attack, defend, collect resources or choose another available game action.

Rewards:

Winning the game → +100

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Completing an important objective → +20

Successful strategic move → +5

Losing resources → -5

Losing the game → -100

Policy:

The policy determines which action the agent should select in each game situation.

Long-Term Strategy

Giving a large reward only for immediate actions may make the agent focus on short-term gains. Rewards for important objectives and winning encourage it to learn long-term strategies.

Exploration

During training, the agent should sometimes try different actions instead of always selecting the same action. This exploration can help it discover better strategies.